



CS683: Advanced Computer Architecture

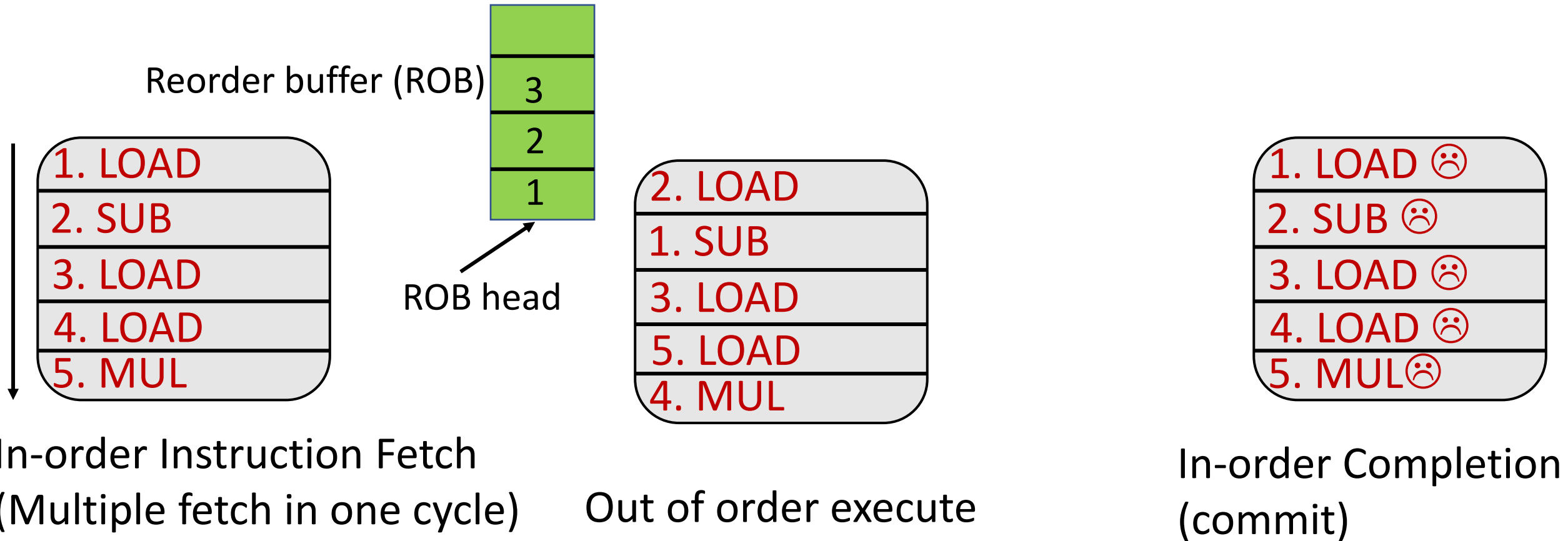
Lecture 17: Monsoon/Thunderstorm is still there 😊

Good-time for Front-end Mysteries

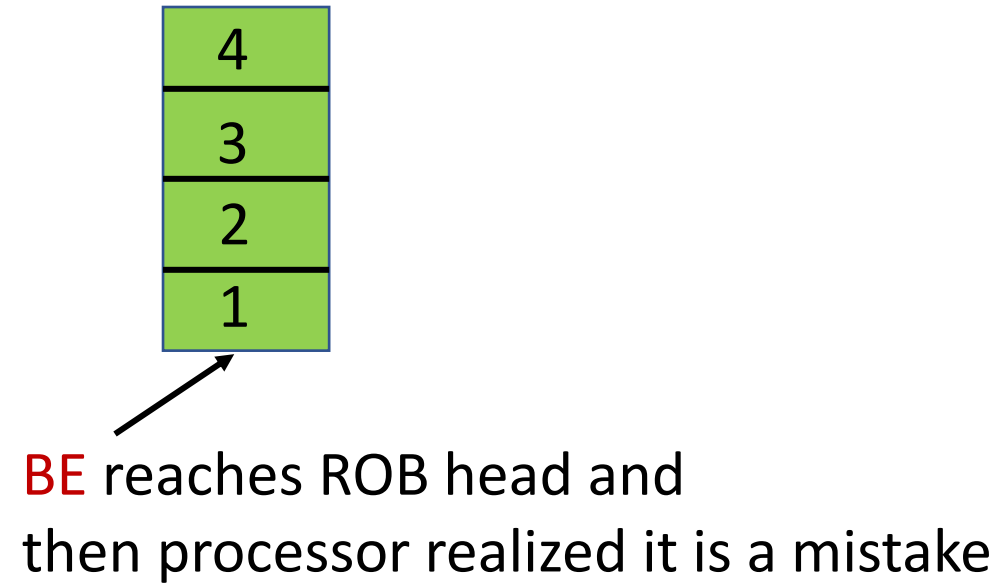
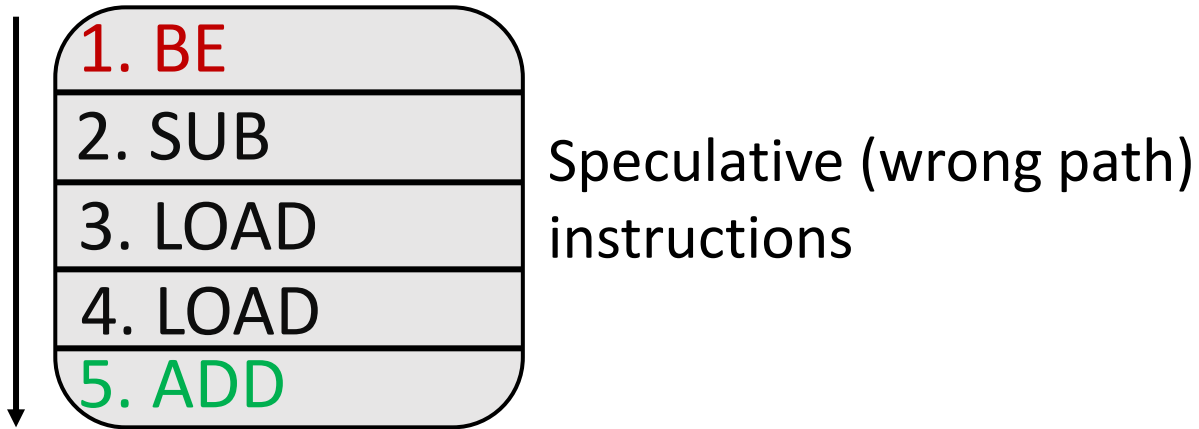
<https://www.cse.iitb.ac.in/~biswa/courses/>

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Modern Processors: In-order Commit



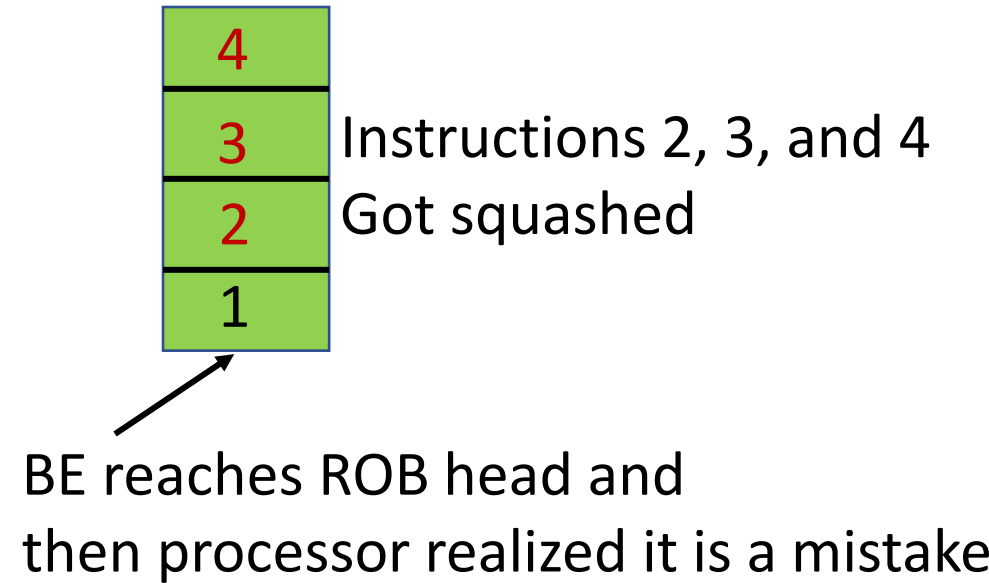
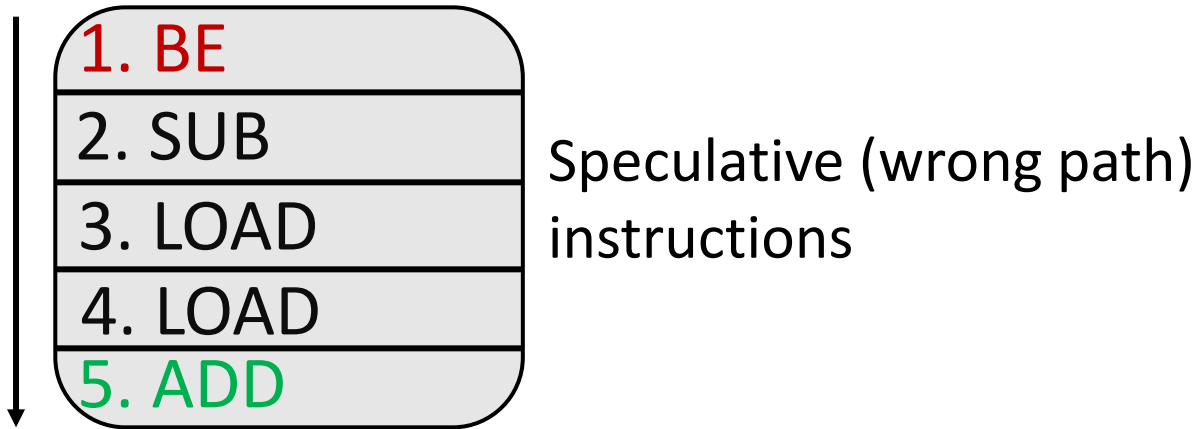
Modern Processors: Speculative Execution



In-order Instruction Fetch
(Multiple fetch in one cycle)

Recent Intel processors have 750-entry ROB
Advanced Computer Architecture

Modern Processors: Speculative Execution



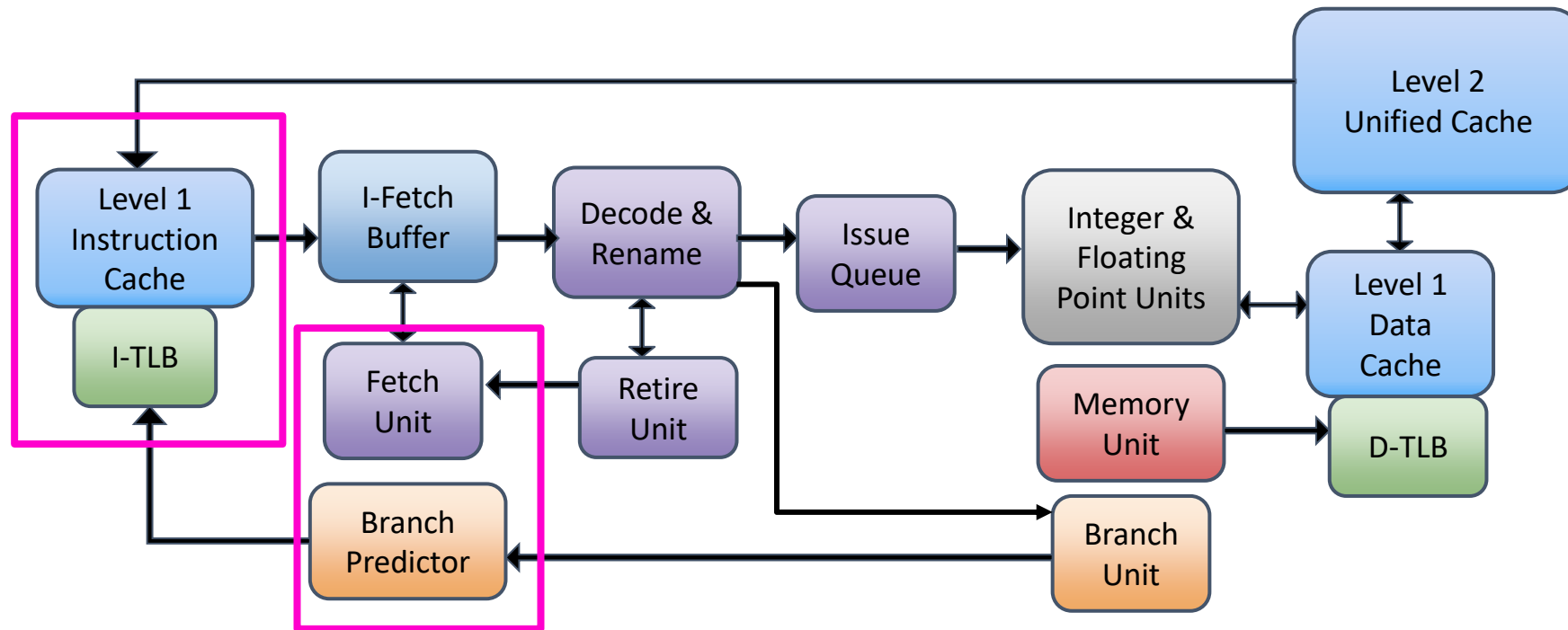
In-order Instruction Fetch
(Multiple fetch in one cycle)

Same happens in the case of a page fault, exception etc...

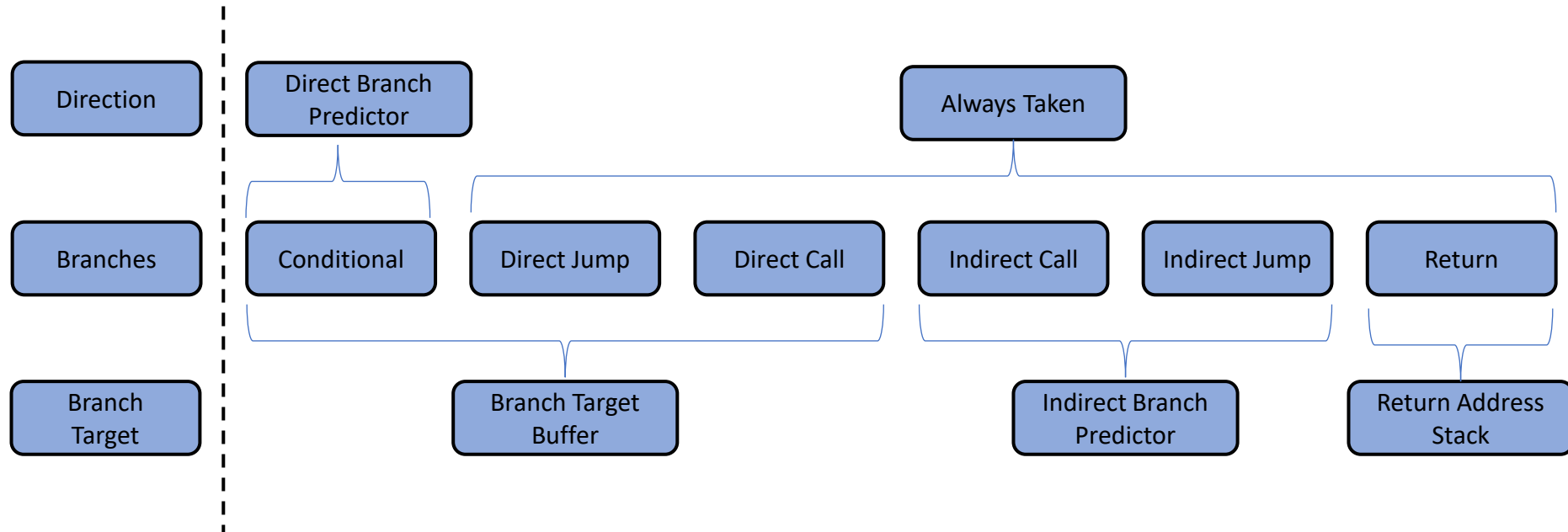
Three Great tastes that tastes great together

- $\text{CPI} < 1$: Go for Superscalar
- Superscalar increases data hazards: Go for out-of-order
- Branch instructions are the problem: Speculative execution

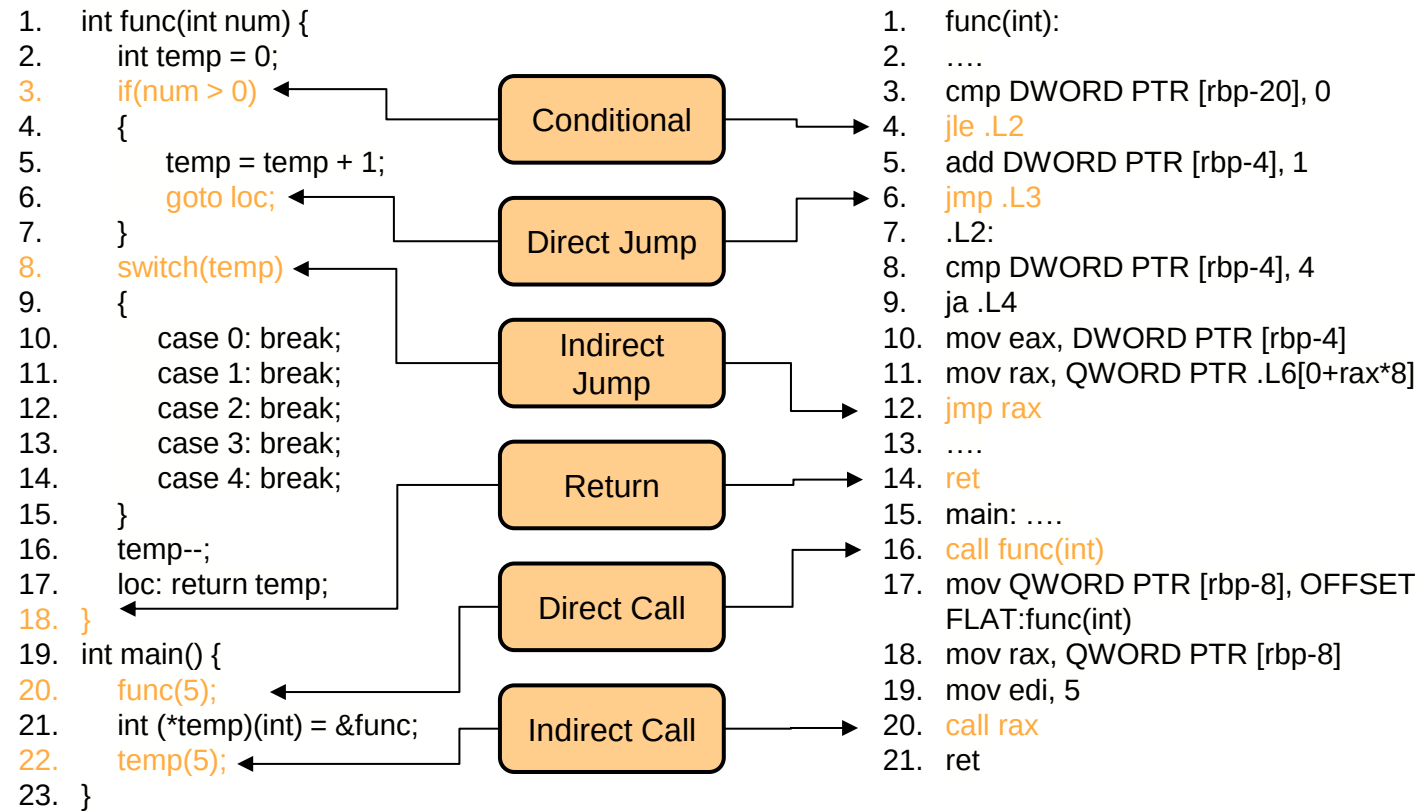
Microarchitecture for Performance



Front-end Bottleneck (Servers, mobile workloads)



Branch instructions



(a) C++ Code

(b) Assembly Code

BTB/L1I Capacity/latency Problem

The AMD “Zen 2” Processor

The branch capacity in Zen 2 is nearly double that of Zen. The L0 BTB was increased from 8 to 16 entries. The L1 BTB was increased from 256 to 512 entries. The L2 BTB was increased from 4096 to **7168 entries**. The indirect target array

The IBM z15 High Frequency Mainframe Branch Predictor

to generation as shown in table 1. The z15 BTB1 can hold up to 16K branches and is organized as 2K logical rows with 8 ways per row. The BTB2 can hold **128K branches** and is organized as 32K logical rows with 4 ways per row.

Evolution of the Samsung Exynos CPU Microarchitecture

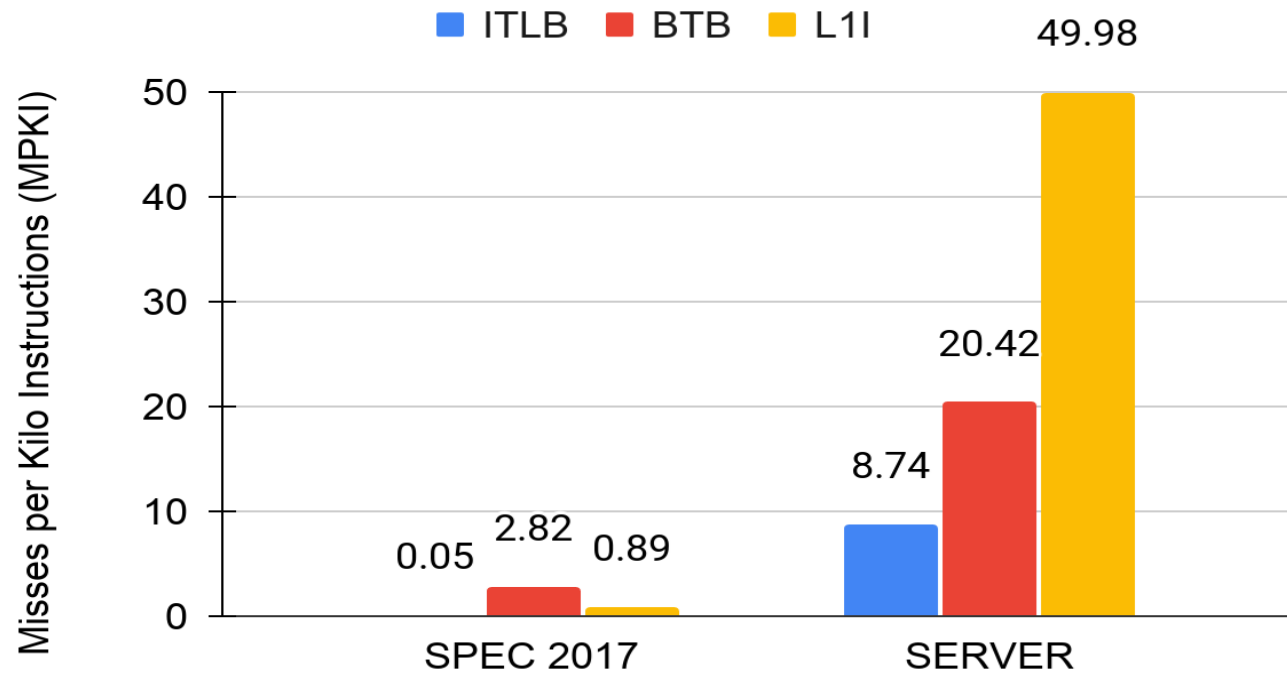
BRANCH PREDICTOR STORAGE, IN KBYTES

Bit storage (KB)	SHP	L1BTBs	L2BTB	Total
M1/M2	8.0	32.5	58.4	98.9
M3	16.0	49.0	110.8	175.8
M4	16.0	50.5	221.5	288.0
M5	32.0	53.3	225.5	310.8
M6	32.0	78.5	451.0	561.5

The Arm Neoverse N1 Platform: Building Blocks for the Next-Gen Cloud-to-Edge Infrastructure SoC

The branch predictor employs a large **6K-**entry main branch target buffer with 3-cycle access latency to retrieve branches' target addresses without accessing the I-cache. Such a

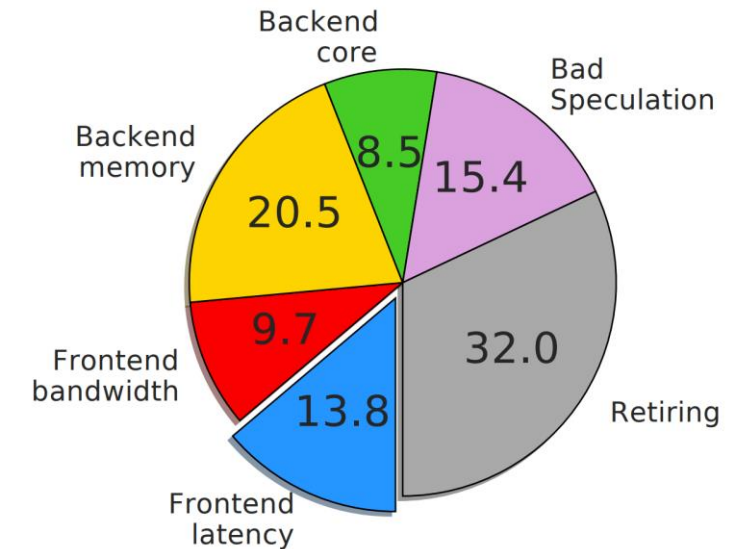
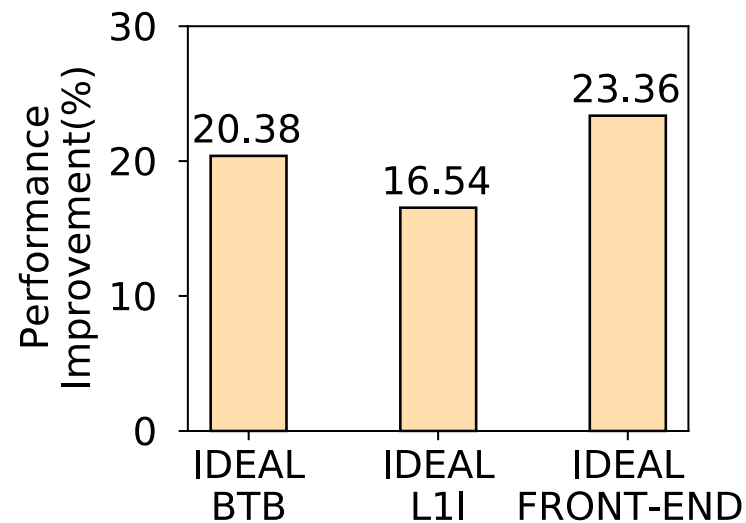
An Example



High MPKI in front-end structures due to increase in codebase.

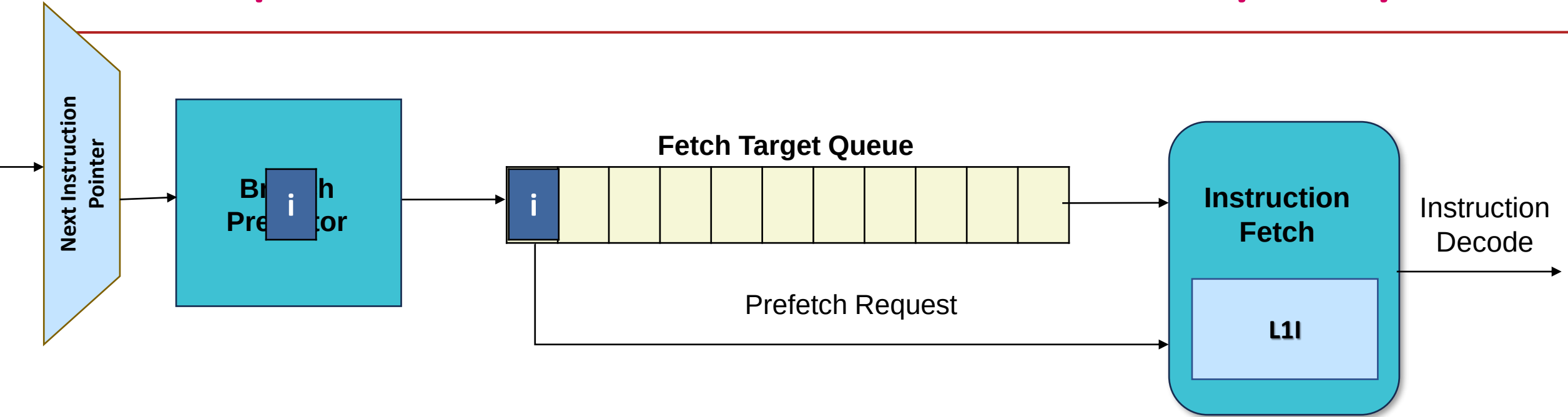
Ideal front-end: Google Web search

Ideal BTB :- All misses converted to hits.
Ideal L1I :- Misses converted to hits.



Ideal front-end does not provide significant performance improvement on top of ideal BTB (it has ideal ITLB too).

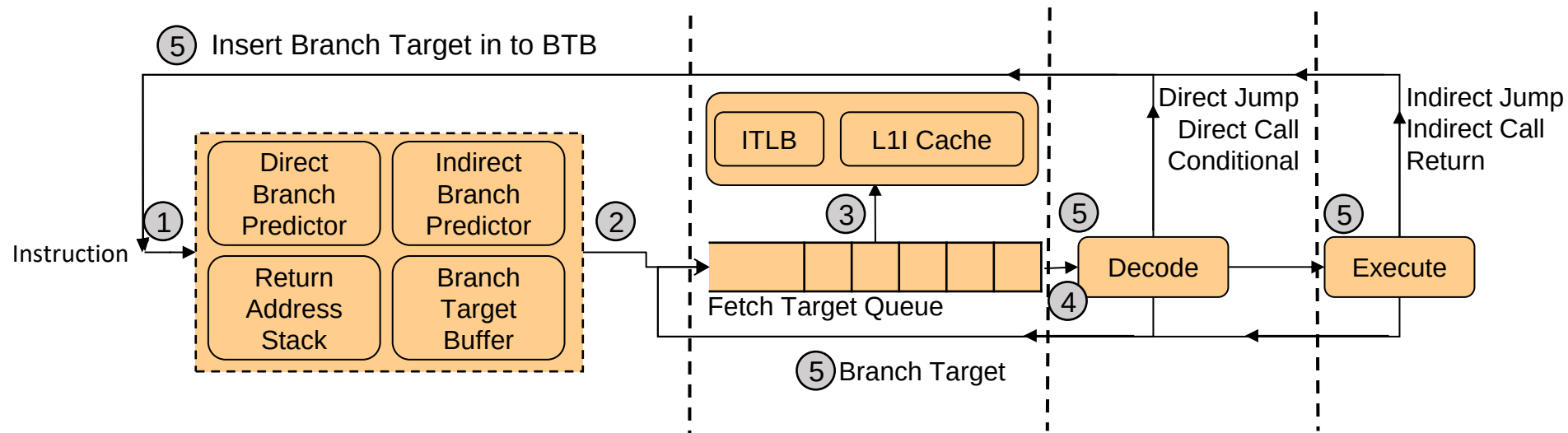
Decoupled Frontend: Welcome to the Mystery



Fetch Directed Instruction Prefetching (FDIP)

Decoupling allows to run way ahead of demand fetch which helps to reduce the impact of long latency L1I misses

Decoupled Front-end



Branch predictor looks ahead even if L1I misses

FDIP paper

<https://cseweb.ucsd.edu/~calder/papers/MICRO-99-FDP.pdf>